Economic Policy Uncertainty and Macroeconomic Dynamics in Latin America*

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Abstract

Numerous studies have analyzed the effect of uncertainty on real macroeconomic variables that drive the business cycle, but a much smaller share have examined its effects in emerging economies. This paper uses a panel vector autoregression (PVAR) model to analyze the effect of economic policy uncertainty in Latin America. The results suggest that, in small open economies, an uncertainty shock has negative effects on the region's economic performance. Greater uncertainty could lead to first-quarter declines of between 1.72% and 2.85% in investment, between 0.49% and 0.65% in consumption, between 0.29% and 0.51% in output, and between 0.05% and 0.17% in employment. These results are consistent with evidence that greater uncertainty is associated with the postponement of investment, consumption, and hiring decisions, dampening economic growth.

JEL Codes: D80, E32, E37, F43, O54.
Keywords: Uncertainty; Growth; GDP; Investment; Employment; Consumption; Latin America.

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1 Introduction

Uncertainty plays a fundamental role in understanding business cycles. It can affect economic agents' confidence in their decisions, leading them to postpone consumption and investment until more information becomes available, with subsequent effects on employment and GDP. Understanding how uncertainty influences these variables is therefore relevant to explaining fluctuations in economic activity.

One mechanism through which uncertainty affects economic activity is the postponement of investment. Most investment expenditures are largely irreversible and can be delayed, allowing firms to wait for new information about prices, costs, and other market conditions before committing their resources (Bernanke, 1983; Pindyck, 1991). Although greater volatility may also encourage investment by increasing potential returns (Kraft et al., 2018), Bloom (2009) shows that uncertainty shocks can lead to a sharp decline in investment and employment, followed by a recovery as uncertainty subsides.

Assessing these effects empirically is challenging because uncertainty is neither observable nor directly quantifiable. An extensive body of literature has therefore developed indicators to measure it and examine its relationship with economic activity (Moore, 2017). Much of this evidence focuses on developed economies. For North America and Europe, de Bondt and Schiaffi (2015) find that confidence helps explain current real GDP growth, while Ghirelli et al. (2021) document negative effects of economic policy uncertainty on private consumption.

In emerging economies, the effects of uncertainty may be larger. Carrière-Swallow and Céspedes (2013) find that uncertainty shocks lead to particularly large declines in investment, together with a substantial decrease in private consumption that is not observed in developed economies. Despite these findings, emerging economies have received less attention in the literature (Cerda et al., 2018). Evidence for Chile also points to the role of external conditions: Fornero and Zúñiga (2019) show that a deterioration in external activity affects expectations and domestic demand, with larger effects on business expectations and investment than on consumer expectations and spending. These results suggest that external conditions are relevant to understanding the relationship between uncertainty and economic activity in the region.

This paper examines the effects of economic policy uncertainty on investment, employment, private consumption, and GDP in a panel of Latin American emerging economies. By studying these economies within a common empirical framework, the analysis complements country-specific evidence for the region. It examines the magnitude and persistence of the responses to an uncertainty shock while accounting for movements in the global business cycle.

To examine these responses, a panel vector autoregression (PVAR) model is estimated using quarterly data from 1997 to 2019, following Abrigo and Love (2016). Uncertainty is measured using the Economic Policy Uncertainty (EPU) index, based on the newspaper coverage methodology proposed by Baker et al. (2016). Given the small open economy characteristics of the countries analyzed, the model also includes principal components constructed from international financial and economic variables to account for external conditions.

The estimates indicate that a one-standard-deviation uncertainty shock is associated with a decline in the growth rates of investment, employment, private consumption, and GDP in the first quarter. The responses of investment, employment, and GDP remain statistically significant for several quarters, whereas the consumption response is significant only in the first quarter. To assess the robustness of these findings, the model is also estimated with Greece and Spain added to the sample. The responses of investment and GDP remain similar, although those of employment and consumption differ from the Latin American results.

The remainder of the paper is organized as follows. Section 2 describes the data, and Section 3 presents the methodology. Section 4 reports the results, and Section 5 presents the robustness check. Section 6 concludes.

2 Data

To analyze the effect of economic policy uncertainty on economic growth in Latin America, this paper uses the *Economic Policy Uncertainty Index* (EPU) proposed by Baker et al. (2016) as a measure of uncertainty. This index seeks to measure uncertainty related to institutional changes by analyzing the frequency of relevant words in newspaper articles. Although the index is available for 28 countries worldwide, the analysis focuses on small open economies in Latin America, specifically Brazil, Chile, Colombia, and Mexico.

The index proposed by Baker et al. (2016) considers terms such as "uncertainty," "economy," "taxes," "deficit," and "congress," among others. These authors developed the EPU indices for Brazil and Mexico. For Chile, the index was developed by Cerda et al. (2018), using monthly data from 1993 onward to count articles containing terms associated with uncertainty, such as "policy," "taxes," "fiscal spending," "public debt," and "congress," among others. For Colombia, the index was developed by Gil and Silva (2019), who included words such as "reform," "deficit," and "corruption," among others.[1]

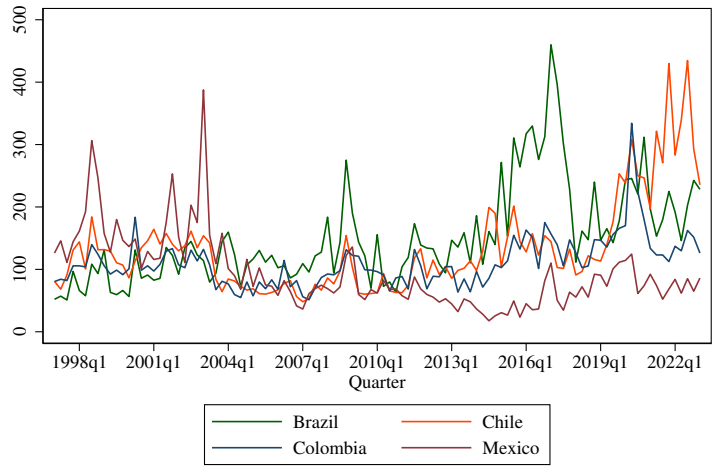
Figure 1 presents the uncertainty indices for each country, showing the quarterly evolution of economic policy uncertainty from the first quarter of 1997 to the last quarter of 2022. At the global level, the effects on uncertainty of the Asian financial crisis in the late 1990s, the dot-com bubble and the Iraq War (Second Gulf War) in the early 2000s, the 2007–2008 financial crisis, and the global COVID-19 pandemic can be observed.

At the country level, notable events in Brazil include corruption scandals, impeachment proceedings, the economic crisis, and various protests between 2014 and 2018 related to the Petrobras corruption scandal, the removal of Dilma Rousseff from office, and cases of contaminated meat in the country. For Chile, notable events include the tax and labor reforms of 2014–2015, as well as the social unrest in late 2019. For Colombia, notable events include the domestic economic crisis of the late 1990s, the political tensions surrounding the proposed referendum in 2000, and the TES (public debt) crisis in 2002, as well as the protests following the announcement of the tax reform bill proposed by Iván Duque's administration in 2021. For Mexico, notable events include the intensification of the Asian financial crisis following the ruble crisis, the increase in the interbank lending rate in August 1998, and the collapse of Latin American stocks and bonds in September of the same year, as well as the September 11, 2001 attacks in the United States and the political and financial problems during Vicente Fox's administration and at PEMEX in 2003.

The growth rates of investment (gross capital formation), private consumption, and output, all measured at constant prices, and the growth rate of employment are used as dependent variables. In addition, following a procedure similar to that outlined by Albagli and Luttini (2015) and Cerda et al.

[1]The methodology was developed by Baker et al. (2016), while the respective authors constructed the country indices following these guidelines. The newspapers used were "Folha de São Paulo" for Brazil; "El Mercurio" and "La Segunda" for Chile; "El Tiempo" for Colombia; and "El Norte," "El Mural," and "Reforma" for Mexico.

Figure 1
Economic Policy Uncertainty Index (EPU).



Note: Author's elaboration based on Baker, Bloom and Davis (2016), Cerda, Silva and Valente (2016), and Gil and Silva (2018).

(2018), several external variables are used to isolate the effect of the global business cycle, given the small open economy characteristics of the countries analyzed.[2]

3 Methodology

Given the nature of the data, a panel vector autoregression (PVAR) model is estimated for the four Latin American economies using quarterly data. This model makes it possible to analyze the impact of various economic shocks that occurred in Latin America between 1997 and 2019, covering periods of heightened uncertainty at both the global and country-specific levels.

Following the methodology proposed by Abrigo and Love (2016) for PVAR analysis, the balanced panel model specified in Equation (1) is estimated.

$$\begin{aligned} Y_{it} &= A_1 Y_{i,t-1} + A_2 Y_{i,t-2} + \cdots + A_k Y_{i,t-k} + B X_{it} + u_i + e_{it} \\ i &\in \{\text{Brazil}, \text{Chile}, \text{Colombia}, \text{Mexico}\}, \quad t \in \{1997q1, \ldots, 2019q4\} \end{aligned} \tag{1}$$

Here, Y_{it} is a vector of endogenous variables, X_{it} is a vector of exogenous variables, u_i is a vector of country-specific fixed effects, and e_{it} is a vector of idiosyncratic white noise disturbances. The

[2] S&P 500, VIX, the federal funds rate, the price of refined copper on the London Metal Exchange (LME), the price of oil (WTI), and world GDP at purchasing power parity.

model is estimated using the Generalized Method of Moments (GMM). Model stability is confirmed by the eigenvalues and their moduli reported in Table A.1 in the appendix, as well as the unit circle plot in Figure A.1 in the appendix. Based on the information criteria reported in Table A.2 in the appendix, a single lag of the variables is included.

Given potential causality issues between the uncertainty index and the real economy, Table A.3 presents the Granger causality test for the model. The results indicate bidirectional Granger causality between the macroeconomic variables and the uncertainty index; for example, economic policy uncertainty Granger-causes GDP growth, and vice versa. The literature offers no definitive answer on this issue: some studies provide evidence that uncertainty causes changes in macroeconomic variables (Ghirelli et al., 2021; Ludvigson et al., 2021), others find causality in the opposite direction (Carriero et al., 2018), and still others find effects in both directions (Angelini et al., 2019). Accounting for this bidirectional relationship is relevant because it could potentially bias the estimates.

The endogenous variables are ordered by decreasing exogeneity, with economic policy uncertainty assumed to be contemporaneously exogenous to the economy as a whole. Thus, the vector of endogenous variables is $Y_{it} = (\text{EPU}_{it}, g^I_{it}, g^L_{it}, g^C_{it}, g^{\text{GDP}}_{it})$, where EPU_{it} is the economic policy uncertainty index and g^I_{it}, g^L_{it}, g^C_{it}, and g^{GDP}_{it} denote the growth rates of investment, employment, private consumption, and output in country i and quarter t, respectively. The Cholesky decomposition follows the same ordering.[3] This ordering implies that shocks to variables placed earlier can contemporaneously affect those placed later, whereas shocks to variables placed later can affect preceding variables only with a lag of at least one period.

Finally, given the small open economy characteristics of the Latin American countries, the international variables described in the previous section are used, following Albagli and Luttini (2015). In line with Cerda et al. (2018), the dimensionality of the matrix of exogenous variables is reduced through principal component analysis[4] to avoid overparameterization. The first three components

[3]This ordering is similar to that used by Cerda et al. (2018), who treated their uncertainty variable as exogenous to shocks in the real economy and placed output last.

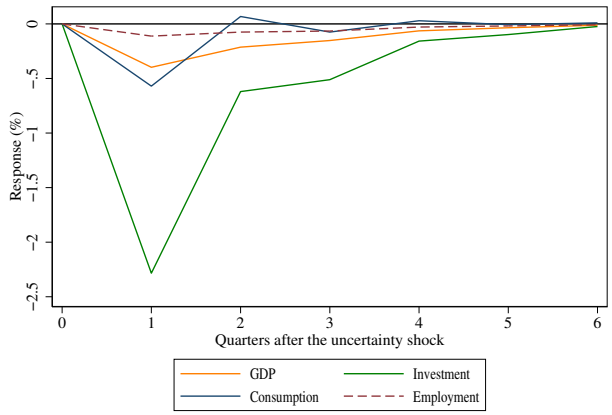
[4]This analysis seeks to capture as much variance as possible in the vectors "PC_i," also called "components," while minimizing information loss. A scree plot, which shows the proportion of variance explained by each component, is used to assess the minimum number of components to retain. In this case, the plot indicates that three components are sufficient to describe the data.

explain 84% of the variance in the international variables, and the vector of exogenous variables is represented by $X_{it} = (PC_1, PC_2, PC_3)$.

4 Results

Following the Cholesky decomposition, Figure 2 presents the orthogonalized impulse response functions for investment, employment, consumption, and output over the first six quarters following the shock, accounting for potential contemporaneous effects among the endogenous variables.

Figure 2
Orthogonalized impulse response functions.



Note: Confidence intervals are omitted to facilitate the visualization of the functions.

For investment, a one-standard-deviation uncertainty shock in Latin America is associated with a negative response in the first quarter, with investment declining by 2.28% and recovering only in the eighth quarter. The 95% confidence interval shown in Figure A.2 in the appendix ranges from -2.85% to -1.72% in the first quarter. The response remains statistically significant through the seventh quarter, providing evidence to reject the null hypothesis of no response.

The results for employment are similar: a one-standard-deviation uncertainty shock in Latin America is associated with a decline of 0.11% in the first quarter. The 95% confidence interval for this period ranges from -0.17% to -0.05%. As in the previous case, the response remains statistically significant, in this case through the tenth quarter.

Consumption also responds negatively to a one-standard-deviation uncertainty shock, declining by

0.57% in the first quarter. The confidence interval for this period ranges from -0.65% to -0.49%. Unlike the previous two cases, the response is statistically significant only in the first quarter following the shock.

The impulse response results for output growth in the countries studied are consistent with the preceding findings. A one-standard-deviation uncertainty shock in Latin America is associated with a negative response in output growth in the first quarter, with output declining by 0.4% and the effect persisting through the ninth quarter. The 95% confidence interval ranges from -0.51% to -0.29% in the first quarter. The negative effect lasts for nine periods, and the response remains statistically significant through the seventh quarter following the shock.

The model results are consistent with the existing evidence and literature, which show that increases in economic policy uncertainty have negative effects on investment, employment, and growth. Although these results do not necessarily establish causality, greater uncertainty is associated with the postponement of investment, consumption, and hiring decisions, which in turn is associated with weaker growth in the Latin American countries analyzed.

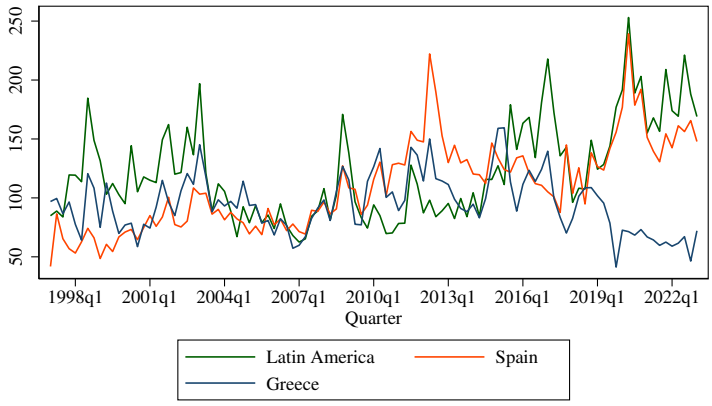
5 Robustness Check

Following the methodology proposed by Abrigo and Love (2016), additional countries are included in the analysis to assess the robustness of the model. This second analysis includes Greece and Spain, whose economic policy uncertainty (EPU) indices are shown in Figure 3.

For Greece, uncertainty increased following the disclosure of the fiscal deficit in 2009 and the subsequent sovereign debt crisis between 2010 and 2012, as well as in response to the general elections and the referendum on the economic bailout held in 2015. For Spain, in addition to the effects of the global financial crisis, uncertainty rose sharply with the Catalan independence process, which began in 2012 and peaked in October 2017, as well as with the 2015–2016 general elections, which were followed by nearly a year without a government. Both countries experienced a slight decline in uncertainty with the adoption of the euro (1999 in Spain and 2001 in Greece).

Using the same PVAR model as in the previous analysis, the additional countries are included in the estimation, and the sample period is shortened by eight quarters relative to the Latin American

Figure 3
Economic Policy Uncertainty Index (EPU).



Note: Author's elaboration based on Baker, Bloom and Davis (2016), Cerda, Silva and Valente (2016), Gil and Silva (2018), Fountas, Karatasi and Tzika (2018), and Ghirelli, Pérez and Urtasun (2019). For illustrative purposes, the figure shows a simple average of economic policy uncertainty for Latin America, based on the four countries analyzed.

sample due to data availability constraints. No other changes are made to the methodology. The model to be estimated is therefore summarized in Equation (2).[5]

$$\begin{gathered} Y_{it} = A_1 Y_{i,t-1} + A_2 Y_{i,t-2} + \cdots + A_k Y_{i,t-k} + BX_{it} + u_i + e_{it} \\ i \in \{\text{Brazil, Chile, Colombia, Mexico, Greece, Spain}\}, \quad t \in \{1999q1, \ldots, 2019q4\} \end{gathered} \tag{2}$$

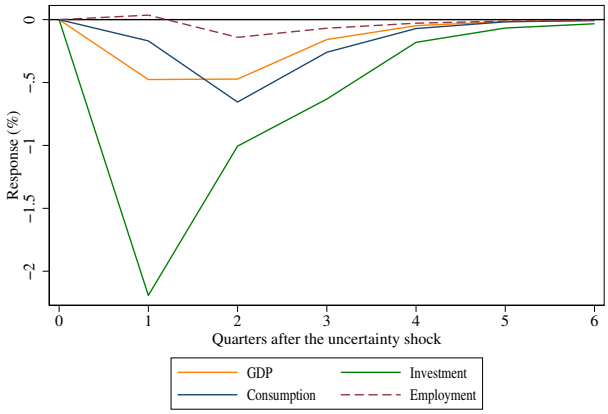
Figure 4 presents the orthogonalized impulse response functions for the sample including the two additional countries, using the same Cholesky ordering as in the previous analysis and one lag, selected based on the information criteria. As the figure shows, the results are visually very similar to those obtained for the Latin American sample.

A one-standard-deviation uncertainty shock is associated with a negative investment response in the first quarter, with investment declining by 2.19%. The 95% confidence interval for this period ranges from -2.47% to -1.9%. In this case, the response is statistically significant in all periods analyzed.

For employment, a one-standard-deviation uncertainty shock is associated with a small positive

[5]The appendix reports the model stability results, information criteria, Granger causality tests, and impulse response functions with their respective confidence intervals.

Figure 4
Orthogonalized impulse response functions.



Note: Confidence intervals are omitted to facilitate the visualization of the functions.

response of 0.04% in the first quarter, with a 95% confidence interval ranging from 0.01% to 0.06%. The response turns negative from the second quarter onward, reaching approximately -0.14% before gradually returning to zero.

Consumption follows a pattern similar to that observed in the initial model estimation, declining by 0.17% in the first quarter, with a confidence interval ranging from -0.24% to -0.1%. The response is statistically significant in all periods analyzed.

The impulse response results for output growth are consistent with the preceding findings. A one-standard-deviation uncertainty shock is associated with a negative response in the first quarter, with output growth declining by 0.48% and the effect persisting through the tenth quarter. The 95% confidence interval for the first quarter ranges from -0.53% to -0.42%. The negative effect persists throughout the ten periods, and the response remains statistically significant through the tenth quarter following the shock.

Despite the inclusion of additional countries, the responses of investment and output remain similar in direction and magnitude to those obtained for the Latin American sample. Consumption also declines, although by a smaller amount, consistent with evidence that consumption is more sensitive to uncertainty shocks in emerging economies (Carrière-Swallow and Céspedes, 2013). By contrast, the employment response changes sign. Overall, these results support the robustness of the main findings for investment and output.

6 Conclusion

This paper examines the effects of economic policy uncertainty on investment, employment, private consumption, and GDP in Latin American emerging economies. Using quarterly data from 1997 to 2019, a panel vector autoregression (PVAR) model is estimated, with uncertainty measured by the Economic Policy Uncertainty (EPU) index and external conditions captured through principal components of international variables.

Within this framework, the results suggest that uncertainty is associated with negative effects on the growth of investment, employment, private consumption, and GDP. In the first quarter following a one-standard-deviation uncertainty shock, investment growth declines by 2.28%, consumption growth by 0.57%, GDP growth by 0.4%, and employment growth by 0.11%. The responses of investment, employment, and GDP remain statistically significant for several quarters, whereas the consumption response is significant only in the first quarter. When Greece and Spain are added to the sample, the responses of investment and GDP remain similar, while the decline in consumption is smaller.

These findings should be interpreted in light of several limitations. Since the EPU index is based on newspaper coverage, its ability to capture the uncertainty faced by economic agents beyond the institutional sphere may be limited. In addition, the analysis is restricted to four macroeconomic aggregates and does not examine the specific channels through which uncertainty affects investment, employment, consumption, and GDP. The bidirectional Granger causality between uncertainty and economic activity also calls for caution when interpreting the estimated responses as causal effects.

Despite these limitations, the negative responses documented in this study suggest that economic policy uncertainty is relevant to understanding fluctuations in economic activity in Latin American emerging economies. The similar investment and GDP responses across the two samples provide further support for this finding. Further research could examine the mechanisms underlying these responses and help explain the differences observed in employment and consumption when the sample is expanded.

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A Appendix

A.1 Latin America

Table A.1
Eigenvalue stability condition.

Eigenvalues			
No.	Real	Imaginary	Modulus
1	-0.44730	0.00000	0.44730
2	-0.19434	0.00000	0.19434
3	0.27499	0.00000	0.27499
4	0.49498	0.11390	0.50792
5	0.49498	-0.11390	0.50792

Table A.2
Model selection criteria.

Lags	CD	J	J p-value	MBIC	MAIC	MQIC
1	0.81	326.04	0.44	-1,564.22	-319.96	-815.32
2	0.02	304.97	0.38	-1,438.99	-291.03	-748.06
3	-4.50	257.85	0.74	-1,339.80	-288.15	-706.84
4	-0.16	190.50	0.98	-1,173.06	-275.50	-632.83

Table A.3
Granger causality tests.

Variable	g^{GDP}	EPU	g^{I}	g^{L}
g^{GDP}	-	0.000***	0.000***	0.000***
EPU	0.000***	-	0.000***	0.000***
g^{I}	0.000***	0.000***	-	0.000***
g^{L}	0.018**	0.000***	0.003***	-

Note: This table reports p-values from Granger causality Wald tests. Each cell reports the p-value for the null hypothesis that the row variable does not Granger-cause the column variable. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Figure A.1
Unit circle, eigenvalues, and model stability.

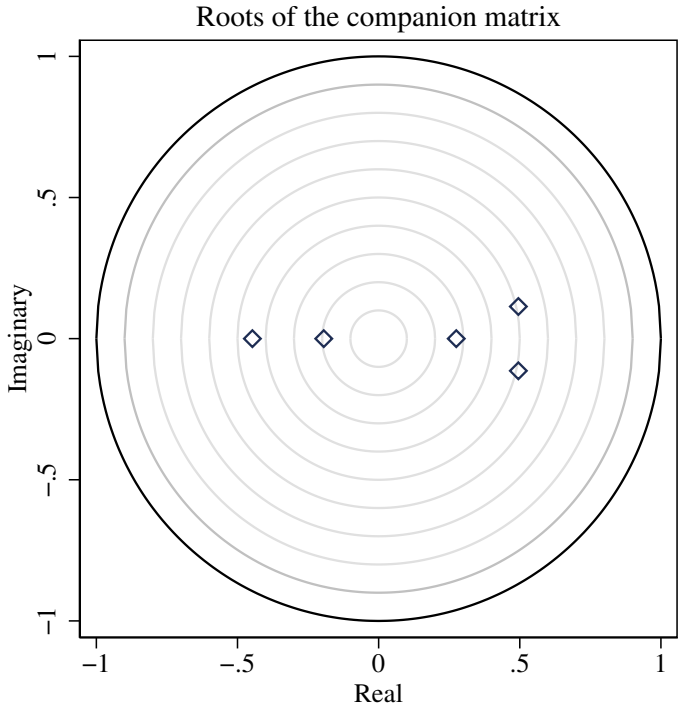
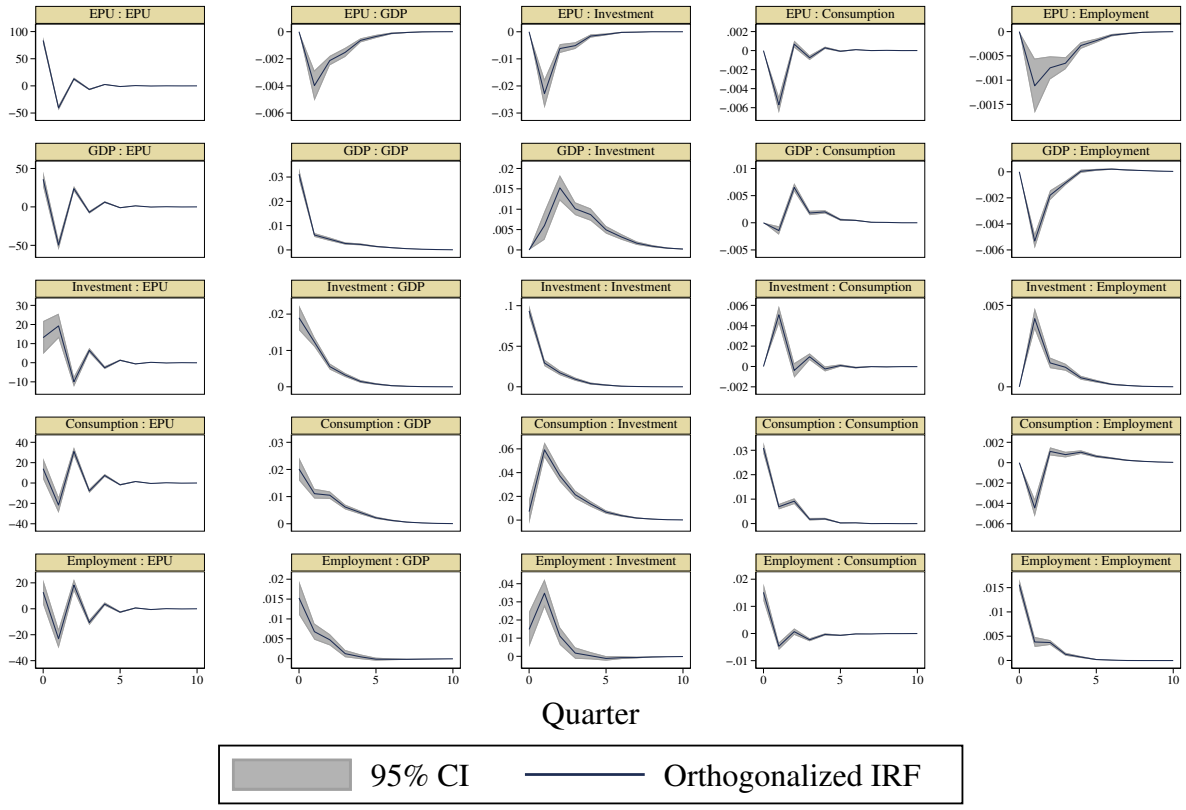


Figure A.2
Orthogonalized impulse response functions.



Note: Each panel title indicates impulse : response.

A.2 Robustness Check

Table A.4
Eigenvalue stability condition.

	Eigenvalues		
No.	Real	Imaginary	Modulus
1	-0.15279	0.11483	0.19113
2	-0.15279	-0.11483	0.19113
3	0.23425	0.24625	0.33987
4	0.23425	-0.24625	0.33987
5	0.55753	0.00000	0.55753

Table A.5
Model selection criteria.

Lags	CD	J	J p-value	MBIC	MAIC	MQIC
1	0.93	444.60	0.55	-2,321.78	-453.40	-1,188.21
2	0.70	433.18	0.37	-2,179.17	-414.82	-1,108.72
3	-0.39	380.08	0.74	-2,078.25	-417.92	-1,070.91
4	-101.19	328.99	0.87	-1,882.88	-389.01	-976.53

Table A.6
Granger causality tests.

Variable	g^{GDP}	EPU	g^{I}	g^{L}
g^{GDP}	-	0.000***	0.000***	0.000***
EPU	0.000***	-	0.000***	0.004***
g^{I}	0.000***	0.000***	-	0.000***
g^{L}	0.000***	0.000***	0.006***	-

Note: This table reports p-values from Granger causality Wald tests. Each cell reports the p-value for the null hypothesis that the row variable does not Granger-cause the column variable. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Figure A.3
Unit circle, eigenvalues, and model stability.

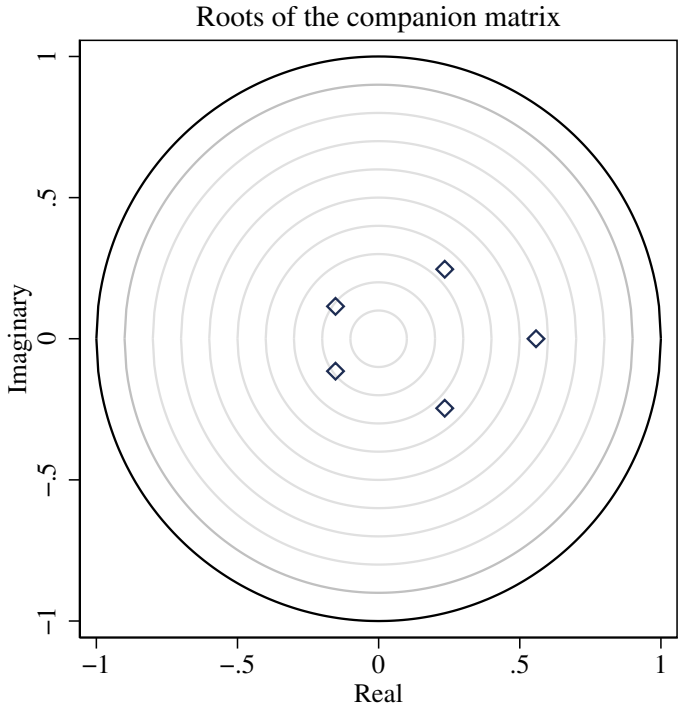
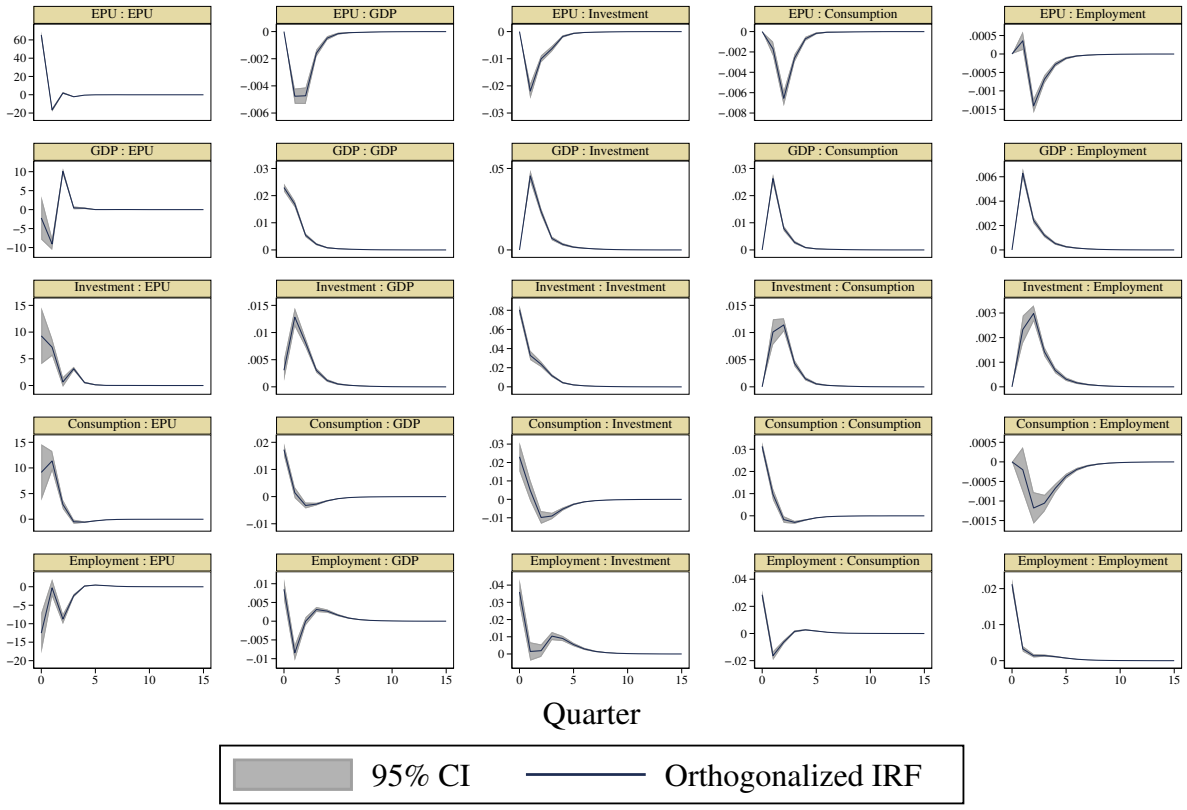


Figure A.4
Orthogonalized impulse response functions.



Note: Each panel title indicates impulse : response.